

- Q.30 Explain the routine maintenance of farm machinery / implements.
- Q.31 Define calibration. And explain why calibrations is necessary?
- Q.32 Write down names of seed metering mechanisms used in planters? Explain any one.
- Q.33 Define zero tillage and strip tillage with their advantages.
- Q.34 Explain the constructional details of seed cum fertilizer drill machine.
- Q.35 Define coupling and their functions.

SECTION-D

- Note:** Long answer type questions. Attempt any two questions out of three questions. (2x10=20)
- Q.36 Describe the constructional details and working principle of disc harrow.
- Q.37 Explain the constructional detail and working principle of paddy transplanter. Also explain preventive maintenance of planters.
- Q.38 Define vibration and its types. Also state their harmful effects.

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Time : 3Hrs.

M.M. : 100

SECTION-A

Note: Multiple choice questions. All questions are compulsory (10x1=10)

- Q.1 An assembly of a number of resistant links connected in such a way that they have relative motion between them.
- a) Machine b) Mechanism
c) Kinematic pair d) Kinematic link
- Q.2 A method of correcting or eliminating unwanted forces
- a) Tension b) Centrifugal force
c) Vibration d) Balancing
- Q.3 Uniformity measure of a planter is
- a) Equal seed spacing
b) Equal number of seed dropped per hill
c) Minimum variation in different openers
d) All of the above
- Q.4 For small distances which drive is used for power transmission
- a) Belt drive b) Rope drive
c) Chain drive d) Gear drive

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- Q.5 The plough used to cut through hard soils by means of a no. of narrow tynes.
 a) Chisel Plough b) M.B. Plough
 c) Disc Plough d) Subsoiler
- Q.6 Disc angle varies from
 a) 20 to 25° b) 25 to 30°
 c) 30 to 35° d) 40 to 45°
- Q.7 The method of planting paddy seeds in field
 a) Transplanting b) Seeding
 c) Broadcasting d) Drilling
- Q.8 Which part of a disc harrow keeps the discs at a fixed position on the shaft?
 a) Gang b) Spool/ Spacer
 c) Bearing d) Scraper
- Q.9 The process of placing the seeds in holes made in seed bed
 a) Drilling b) Broadcasting
 c) Dibbling d) Seeding
- Q.10 Which power source is required to operate a rotavator?
 a) Hydraulic System
 b) Draw bar power
 c) PTO (Power take off)
 d) Stationary engine

SECTION-B

Note: Objective type questions. All questions are compulsory. (10x1=10)

- Q.11 Write down any two type of harrows.

- Q.12 name any two plough accessories.
- Q.13 Give classification of farm machines according to hitching.
- Q.14 Define secondary tillage implements.
- Q.15 Write down types of vibrations.
- Q.16 Dynamometer is used for _____.
- Q.17 Write types of joints used in agriculture machinery.
- Q.18 Define structure.
- Q.19 Define Draw bar.
- Q.20 Give any two examples of primary tillage implements.

SECTION-C

Note: Short answer type questions. Attempt any twelve questions out of fifteen questions. (12x5=60)

- Q.21 Define farm mechanization. Also state any three scopes of farm mechanization.
- Q.22 Write down advantages of rotavator over other tillage implements.
- Q.23 What are different methods of sowing? Explain any one method.
- Q.24 Describe the working principle of potato planter.
- Q.25 What are different types of kinematic pair?
- Q.26 Explain the different types of furrow openers.
- Q.27 Describe the working principle of hydraulic system.
- Q.28 Classify farm implements according to hitching.
- Q.29 Write down the difference between seed drill and planter.